Telecom Customer Churn Prediction & Retention Strategy

Keeping the right customers at the right moment: an ML-guided, test-first retention programme for an anonymised telecom operator

Data science portfolio project · analytical case study, not a production deployment

Executive summary

Churn risk peaks at a predictable moment. ML finds who is at risk; a pilot proves which low-cost actions pay

65.1%

churn at month 11 of tenure, vs 28.8% at month 9 [D]

1.49×

more churners in the model's riskiest 20% than in a random 20% (test ROC-AUC 0.695) [D]

−$328k

loss per monthly campaign if every flagged customer gets a $60 discount [A]

≈19%

uplift a $10 action must achieve to break even — to be measured in a pilot [A]

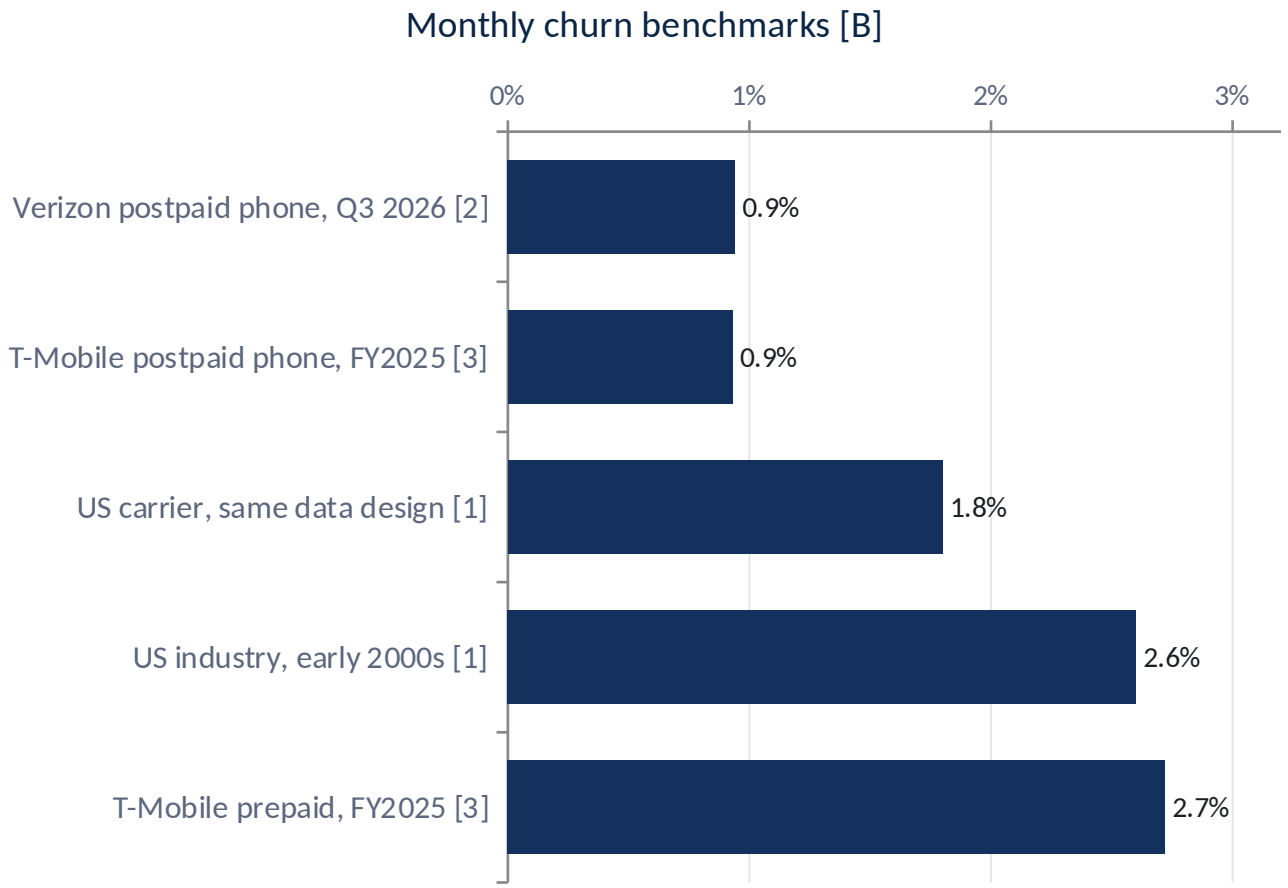
Recommendation

Score customers monthly → focus on high-value customers at risk around the renewal window → test low-cost actions in a randomised pilot → scale only actions whose measured uplift beats break-even

[D] dataset finding · [A] scenario: 2% monthly churn, per 100,000 customers, per monthly campaign · details on slides 5, 8, 11 and 13

Retention is a profit lever

Customers are costly to replace and margins are high — but only well-targeted retention pays



21.5% a year

of customers lost at 2% monthly churn ($1 - 0.98^{12}$, constant-rate assumption) [A]

$300–700 vs $60–100

estimated cost to replace a lost customer [1] vs high-end retention incentive [5] [B]

45–48%

EBITDA margin at large US operators: Verizon Consumer 45.3% [2], T-Mobile 48% of service revenue [4] [B]

Sources: [1] Lemmens & Croux 2006 (replacement cost cites Snel 2000) · [2] Verizon Q3 2026 8-K · [3] T-Mobile FY2025 results · [4] T-Mobile 10-K FY2025 · [5] RCR Wireless / RBC 2003

Data and approach

100,000 customers, one question: who will leave in the next 31–60 days, and what is it worth to act?

100,000 customers

Account table + usage table joined one-to-one on Customer ID; ~100 variables; no duplicates

≈1 month to act

Label = customer left 31–60 days after the snapshot, so a prediction gives time to intervene

49.6% churn — by design

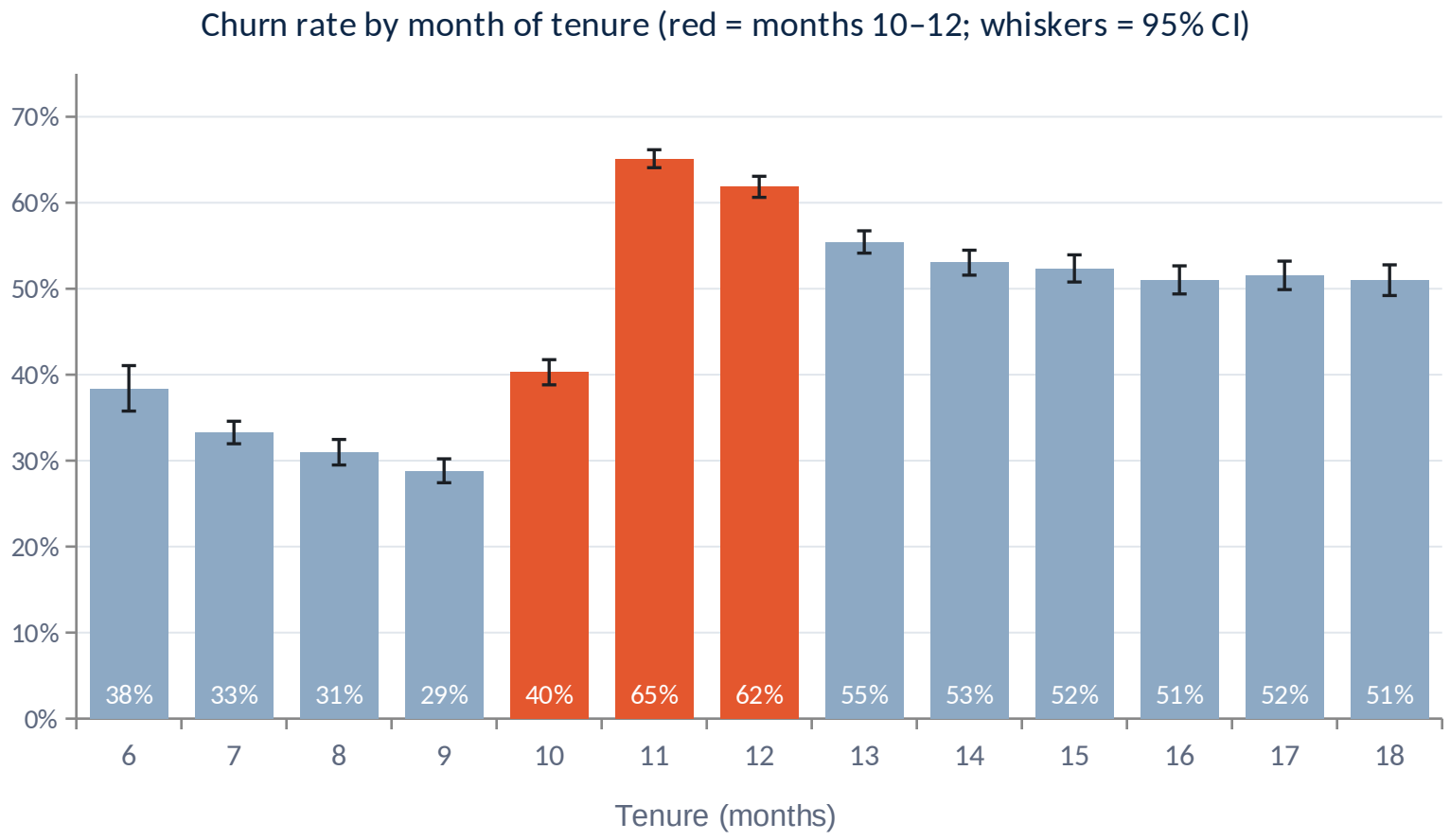
The sample was balanced on purpose. Real churn is far lower, so 50% is never used as Company A's rate

Explore	Validate	Predict	Segment	Cost & test
Find patterns in churn	Test them statistically	Score each customer's risk	Group by cause and value	Break-even + pilot design

[D] Company A dataset (Client.csv + Record.csv). The 49.6% churn rate is a sampling choice; all financial scenarios use benchmark-based real churn of 1–2.7% per month

Churn risk jumps around month 11

Risk more than doubles from month 9 to month 11 — and never returns to its earlier low



28.8% → 65.1%

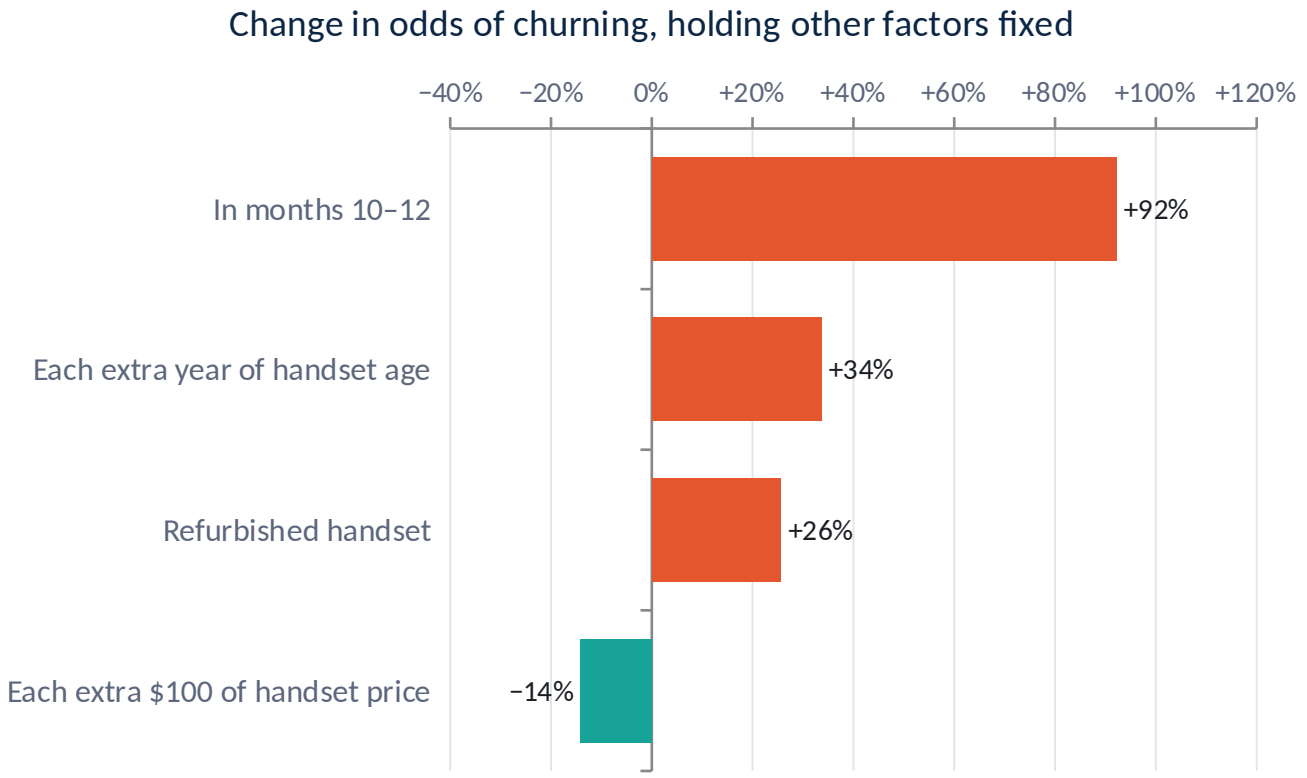
month 9 → month 11

- Whiskers = 95% confidence intervals (Wilson method, as in the notebook). Month 9: 27.4%–30.2%; month 11: 64.1%–66.2% — they do not overlap.
- After month 12, churn settles at ~51–55% — above the pre-month-10 level.
- Consistent with a 12-month commitment ending, but there is no contract field: we call it a risk window.

[D] Company A data, churn rate by tenure month (6–18); error bars = 95% Wilson confidence intervals. Sample churn ≈50% by design: rates compare groups, not real churn levels

The pattern holds after controls

Months 10–12 and handset age raise churn odds even after accounting for revenue, usage, call quality and more



Red = raises churn odds · Green = lowers churn odds

Stable
Months 10–12 odds ratio as controls are added: 1.87 → 1.89 → 1.92

Independent support
A study of a same-design US carrier found churn rising in a handset's first year, linked to 1-year contract + free-phone packages [1]

Rejected
Risk does not fall back after month 12 — the shape is a step up, not a temporary spike

These are associations: they show when risk peaks, not what would stop it.

[D] Logistic regression on 98,269 customers, controlling for tenure trend, handset price, revenue, minutes, usage change, care calls, call failures and refurbished handsets. [B] [1] Lemmens & Croux 2006

The prediction model

We compared two model types fairly and chose the one that ranks risk best without memorising the data

Model	CV ROC-AUC	Overfitting gap*	Top-20% lift
Logistic Regression (baseline)	0.655	0.005	1.38×
LightGBM (default)	0.695	0.112	1.47×
LightGBM (conservative) — selected	**0.694**	**0.067**	**1.47×**

*Training AUC minus validation AUC: smaller = less memorisation

Selected model: LightGBM (conservative)

Metric: ROC-AUC · 5-fold CV score 0.694 ± 0.004 · Test score 0.695

Why: beats Logistic Regression in all 5 folds, captures the month-11 step and interactions, same accuracy as the default LightGBM with about half the overfitting gap.

How it works

Small decision tree #1

Tree #2 corrects #1's errors

… 800 small trees in total

→ churn-risk score for every customer

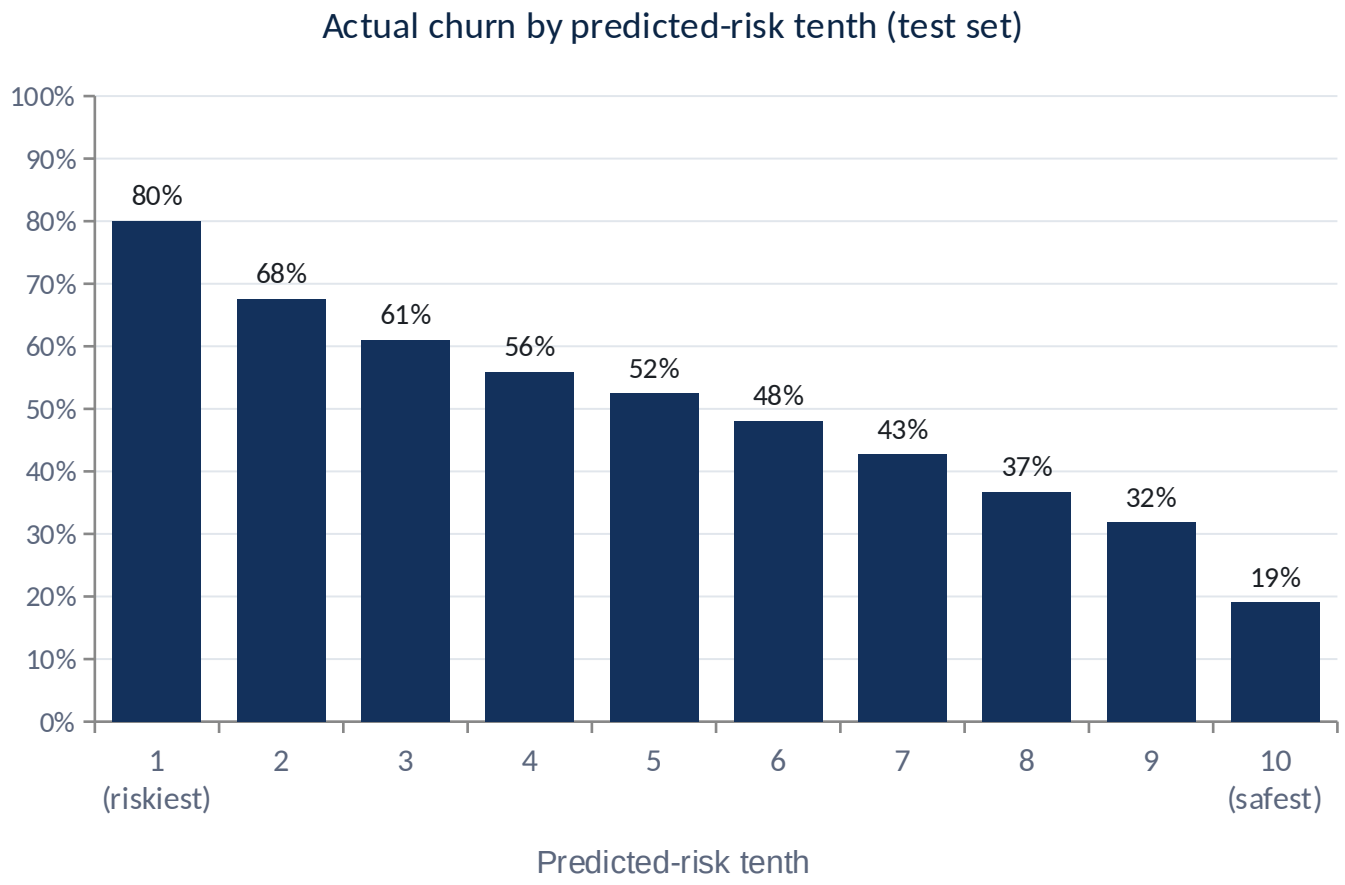
Safeguards

- Data preparation learned inside each training fold only
- No leakage found: best single variable scores only 0.57 AUC
- Test set used once, after all choices were locked

[D] 5-fold stratified cross-validation on 80,000 training customers; 20,000 test customers held out. ROC-AUC = how often the model ranks a real leaver above a stayer (0.5 = coin flip, 1.0 = perfect)

Performance on unseen customers

On 20,000 held-out customers, results matched cross-validation — the riskiest tenth churn at 80% vs 19% for the safest



Metric (LightGBM conservative)	Test	CV mean
ROC-AUC (ranking quality)	**0.695**	0.694
PR-AUC	**0.683**	0.679
F1-score (threshold 0.5)	**0.638**	0.638
Precision, riskiest 20%	**73.8%**	72.9%
Recall, riskiest 20%	**29.8%**	29.4%
Lift, riskiest 20%	**1.49×**	1.47×

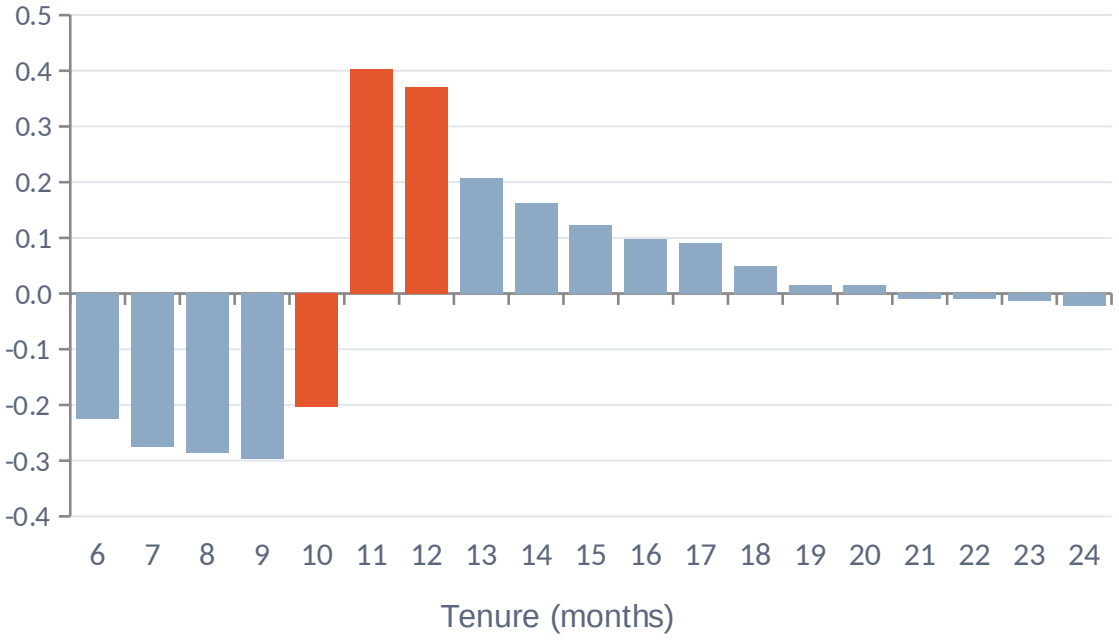
Accuracy alone would hide which mistakes matter: a missed leaver and a wasted offer cost different amounts.

[D] Test set, LightGBM (conservative). Rates come from the balanced 50% sample; at realistic churn rates absolute precision is much lower (slide 11), while ranking quality (AUC, lift) carries over

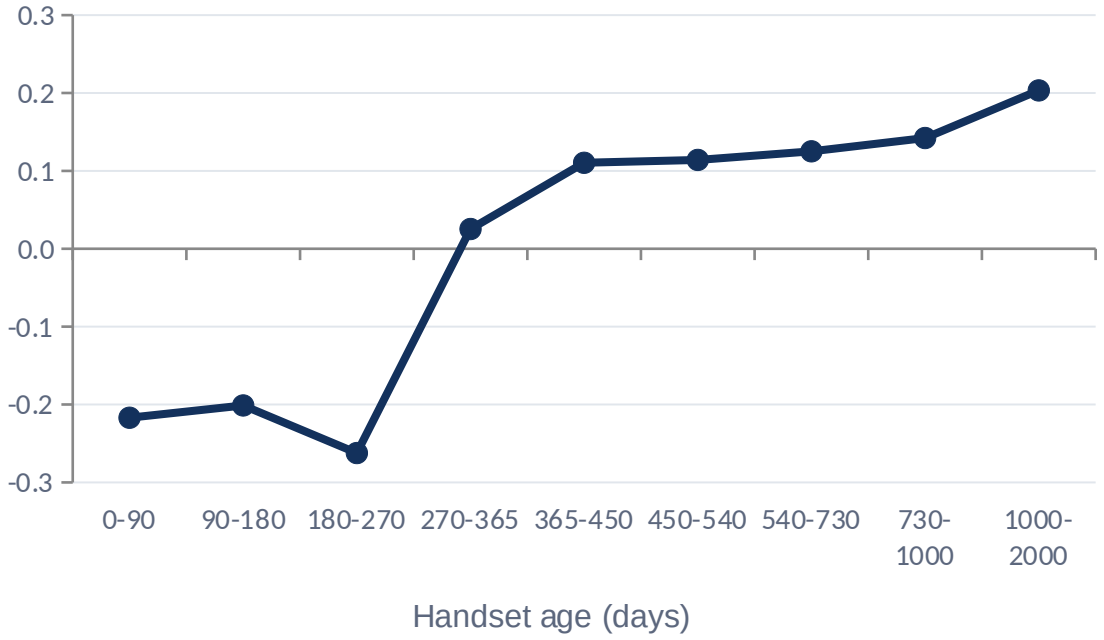
What drives predicted risk

The model independently rediscovered both findings: a month-11 spike and a jump once handsets pass about one year

Effect of tenure on predicted risk (red = months 10–12)



Effect of handset age on predicted risk



Top drivers: handset age (days), change in minutes vs 3-mo avg, tenure (months), revenue per minute. Example: a customer at month 11 whose usage fell 123 minutes below their own average → predicted risk 0.90 (balanced-sample scale).

[D] SHAP values on the test set (log-odds; above 0 = pushes predicted risk up). SHAP explains what the model uses — it is not proof of what would change behaviour

Different risks need different actions

Among the riskiest 20% of test customers, groups differ in cause and value — so no single discount fits

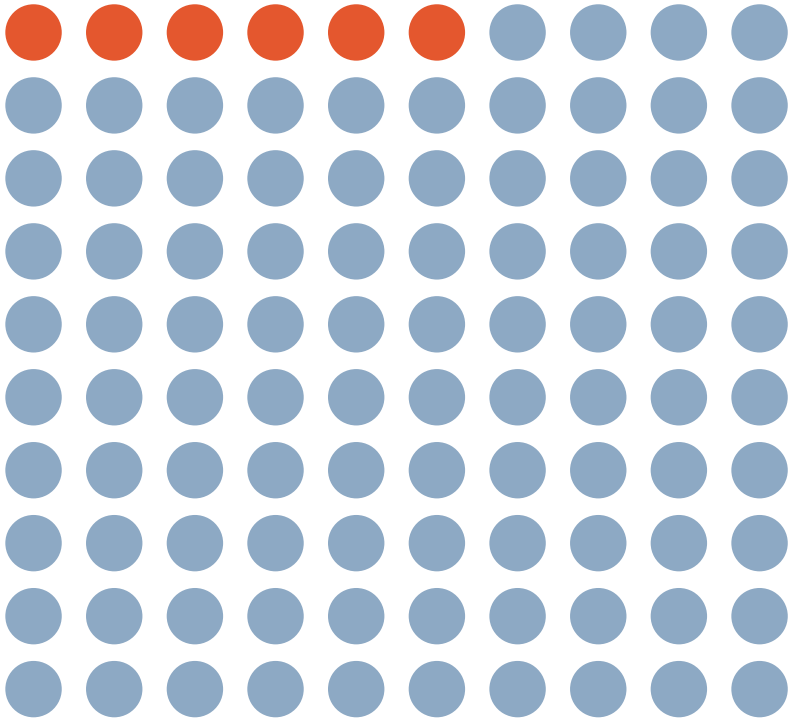
Segment	Customers	Sample churn	Avg revenue / month	Action to test
Data-quality review	40	82.5%	$65	Fix the data and re-score — missing data does not mean inactive
Likely inactive	281	83.6%	$24	Cheap win-back check only; do not spend heavily
Renewal window	1,352	75.9%	$60	Pre-renewal outreach at months 9–10
Ageing handset	1,906	71.1%	$53	Customer-financed upgrade with a new commitment
Service quality	133	78.2%	$74	Network fix or service credit, not a discount
Other high risk	288	69.4%	$84	Investigate; general retention

Risk is not value: the riskiest group (inactive, 84% churn) brings only $24/month, while smaller groups carry the most revenue. Highlighted: missing usage data reflects a data gap — these customers show normal recent revenue.

[D] Riskiest 20% of the balanced test sample (4,000 customers); sample churn rates (~50% base), so compare groups, not levels. Segments assigned in the order listed

Why blanket discounts fail

A $60 offer to every flagged customer loses ≈$328k per monthly campaign — about 94% of them would have stayed anyway



- 6 in 100 would actually leave
- 94 in 100 would stay anyway — but cost money



Customers contacted
10,767

Would actually leave
596 (5.5%)

Net benefit
−$328,445

ROI
-95%

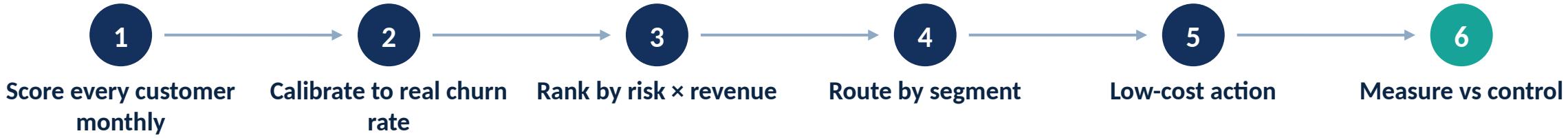
Break-even uplift
214% — impossible

The model list is still 2.8× richer in churners than a random list — but that cannot rescue a blanket discount.

[A] Design A, medium scenario: 2% monthly churn, 100,000 customers, $60 offer [5], 50% acceptance, 10% uplift, $2 contact, 45% margin [2], 12-month horizon. Benefit counts contribution margin of incremental saves only

A better strategy: value and timing

Rank by calibrated risk × revenue, act around the renewal window, and use low-cost or self-funding actions



	A: blanket discount (rejected)	B: value-focused, low-cost (proposed)
Who is contacted	Everyone above a risk cut-off	**Top 2% by calibrated risk × revenue**
Customers per month	10,767	**2,002**
Value of one saved customer	$270	**$688**
Cost per accepted offer	$60 (historical credit)	**$10 (low-cost action)**
Break-even uplift	214% — impossible	**18.8% — testable**

Even on the value-focused list, historical credits ($35–60) would need 52%–86% uplift to break even — so we avoid them.

[A] Medium scenario. Calibration: model probabilities (learned on a 50% sample) are adjusted to the assumed real churn rate before multiplying by revenue — this assumes the sample differs from reality only in its churn proportion. Offer costs [5]

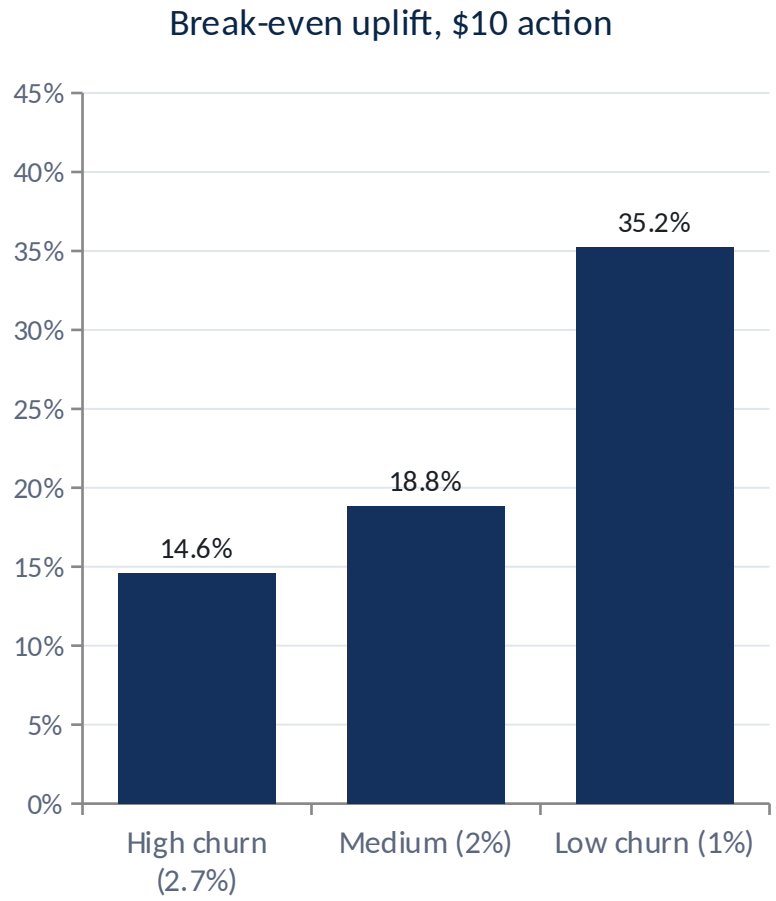
Financial feasibility

With a $10 action the programme breaks even at ≈19% uplift — the pilot must show whether that is achievable

Net benefit per monthly campaign (green = pays off)

Uplift / Cost	$0	$5	$10	$20	$35	$60	$100
5%	−$0.3k	−$5.3k	−$10.3k	−$20.3k	−$35.3k	−$60.4k	−$100.4k
10%	+$3.4k	−$1.6k	−$6.6k	−$16.6k	−$31.6k	−$56.6k	−$96.7k
15%	+$7.2k	+$2.2k	−$2.9k	−$12.9k	−$27.9k	−$52.9k	−$93.0k
20%	+$10.9k	+$5.9k	**+$0.9k**	−$9.1k	−$24.2k	−$49.2k	−$89.2k
30%	+$18.3k	+$13.3k	+$8.3k	−$1.7k	−$16.7k	−$41.8k	−$81.8k
40%	+$25.8k	+$20.8k	+$15.7k	+$5.7k	−$9.3k	−$34.3k	−$74.4k

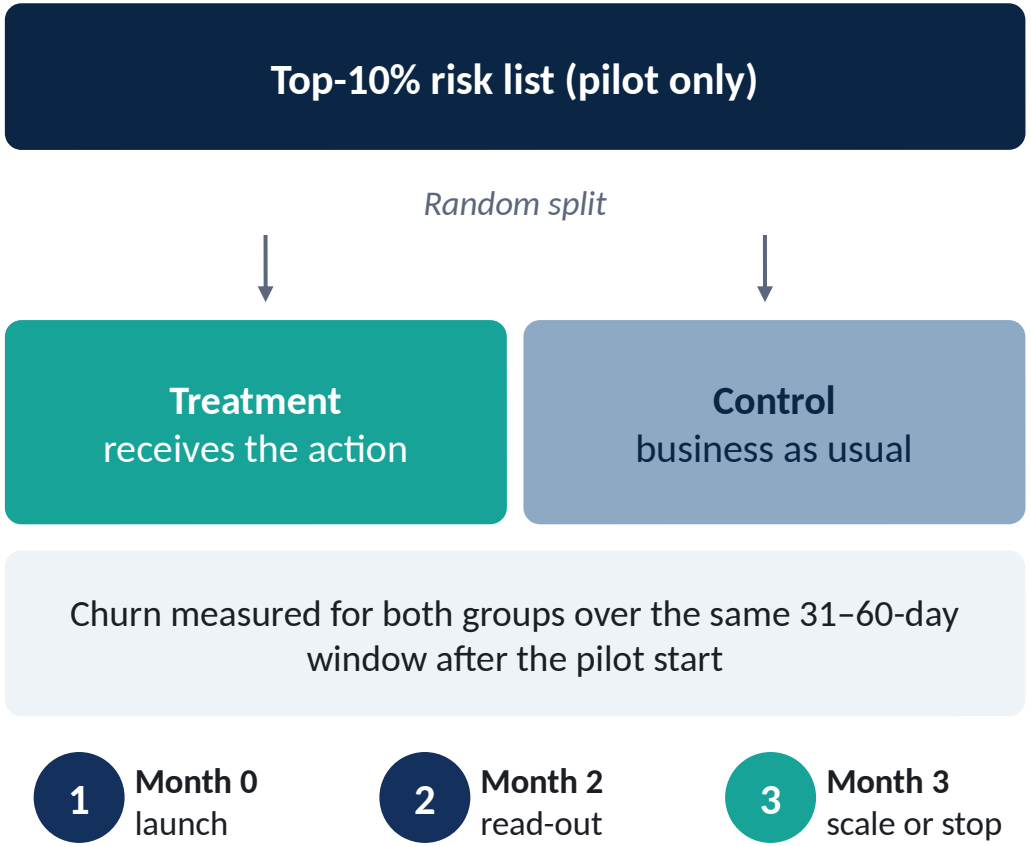
Illustrative scenario, not a guaranteed result: at 20% uplift and a $10 action, net benefit is $865 per monthly campaign (ROI 6%) — essentially break-even.



[A] Design B, per monthly campaign, 100,000 customers; medium: 2% churn, 45% margin [2][4], 50% acceptance, $2 contact, 12-month horizon. Uplift = share of would-be churners saved; acceptance and contact cost are illustrative. Offer costs [5]

Proving it: a randomised pilot

Measure the effect before spending at scale — and scale only actions whose measured uplift beats break-even



Two lists, two purposes

- Top-10% risk list: only to measure uplift quickly — 5,952 per group ≈ 1.2 months of lists
- Calibrated value top-2% list: the scale-up strategy — measuring there would need 6,218 per group ≈ 6 months

Measured separately

- Acceptance = acceptors ÷ customers offered → drives cost
- Uplift = (control churn − treatment churn) ÷ control churn, everyone randomised counted → drives benefit
- Complier-effect estimate = churn reduction ÷ acceptance (effect among acceptors; valid only under the assumptions below)

Decision rule

- Scale only if measured uplift exceeds break-even for the action's real cost — and holds for high-value segments

[A] Sample sizes: two-sided test, 5% significance, 80% power, 2% monthly churn, detecting a 20% relative churn reduction; cross-checked with statsmodels. Complier estimate assumes: random assignment, no offer in control, offer works only through acceptance

Value, limitations and references

Value comes from avoiding a costly mistake now — and earning a measured return only if the pilot proves it

Business value

- Avoids ≈$328k loss per monthly campaign vs a blanket discount [A]
- If the pilot shows 30% uplift with a $10 action: ≈+$8.3k net per monthly campaign per 100,000 customers — illustrative [A]

Limitations

- Balanced sample: probabilities re-calibrated by assumption, not verified on real data
- No time-based validation; cause of the month-11 spike unknown
- Uplift, acceptance, contact cost and horizon are assumptions
- Offer-cost (2003) and same-design churn (2006) sources are old

References

[1] Lemmens, A. & Croux, C. (2006). Bagging and boosting classification trees to predict churn. Journal of Marketing Research, 43(2), 276–286. https://scispace.com/pdf/bagging-and-boosting-classification-trees-to-predict-churn-boq2zc7149.pdf

[2] Verizon Communications (2026). Q3 2026 earnings release, Form 8-K Exh. 99. https://www.sec.gov/Archives/edgar/data/0000732712/000073271226000049/a2026q3exhibit99.htm

[3] T-Mobile US (2026). Q4 & FY2025 results (press-release reproduction). https://quantisnow.com/insight/t-mobile-delivers-best-in-class-customer-results-in-q4-translating-into-durable-6385633

[4] T-Mobile US (2026). Form 10-K FY2025. https://www.sec.gov/Archives/edgar/data/1283699/000128369926000010/tmus-20251231.htm

[5] Gregson, R. (2003, Aug 18). Carriers step up retention efforts (RBC Capital Markets survey). RCR Wireless News. https://rcrwireless.com/20030818/archived-articles/carriers-step-up-retention-efforts

[6] T-Mobile US (2026). Form 10-Q Q3 2026. https://www.sec.gov/Archives/edgar/data/0001283699/000128369926000129/tmus-20260930.htm

Data: anonymised telecom customer dataset (not redistributed). Full analysis: notebooks/telecom_churn_analysis.ipynb